

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Bioscience and Biotechnology
2. **Subject Name:** Advances in Protein Structure and Function **L-T-P: 3-1-0**
Credits: 4
3. **Pre-requisites:** Biochemistry for UG students, nil for PG and PhD students
4. **Compulsory or Elective:** Compulsory for UG, elective for PG and PhD level
5. **Level of the subject:** UG, PG and PhD Level
6. **Syllabus and reference books:**

Syllabus:

Ionization of amino acids and peptide, determination of pI; Protein folding problem, Anfinsen's experiment; Motifs and domains of proteins; Protein evolution; Molecular recognition, thermodynamics and specificity of ligand binding; Allostery.

Nascent protein conformation and its interaction with the surroundings; Knowledge based protein designing and selected case studies; Directed protein evolution: Phage display, cell surface display, cell free display system; mutagenesis studies; alternative scaffolds.

Reference Books:

Introduction to Protein Science: Architecture, Function, and Genomics (2010) by Arthur M Lesk, Oxford University Press.

Introduction to Protein Structure (1998) by Branden & Tooze by Garland Publishing.

Protein Engineering and Design (2010) by Sheldon Park and Jennifer Cochran, CRC Press.

Molecules of Life (2013) John Kuriyan, Boyana Konforti and David Wemmer, Garland Science.

7. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Ionization of amino acids, zwitter ion, determination of pKa and pI of amino acids	2
2.	Titration of peptides and determination of pI	2
3.	Forces that stabilize protein fold, Anfinsen's experiment	2
4.	Motifs and domains of proteins	2
5.	Mechanisms of protein evolution, study of evolution by comparison of proteins sequences and structures, BLOSUM62 matrix	4
6.	Affinity of protein-ligand interaction, thermodynamics of protein-ligand interaction, experimental methods to determine affinity of interaction	3
7.	Specificity of molecular recognition, hormone-receptor interactions, protein-nucleic acid interactions	3
8.	Allostery, case study of hemoglobin	2
9.	Nascent protein conformation	3
10.	Nascent protein and its interaction with the surroundings	2
11.	Knowledge based protein designing and selected case studies	6
12.	Directed protein evolution: Phage display	2
13.	Directed protein evolution: Cell surface display and cell free display system	3
14.	Mutagenesis studies; alternative scaffolds and case studies	4
15.	Tutorial: Various problems/exercises will be assigned on the structure-function relationship of proteins taught in the class. These problems will be discussed and solved. Use of Pymol, Discovery Studio and other programs will be taught.	10
Total number of hours		50